Question	Answer	Marks
4(a)	light diffracts / spreads (at slit(s))	**B1**
	light (from each slit) superposes / interferes (at screen)	**B1**
	when phase difference is 0 / path difference is $n\lambda$ (where n is an integer) a bright fringe is formed **or** when phase difference is 180(°) / path difference is $(n + 1)\lambda / 2$ (where n is an integer) a dark fringe is formed	**B1**
4(b)	$\lambda = ax / D$	**C1**
	$\lambda = 1.0 \times 10^{-3} \times 3.3 \times 10^{-3} / 4.8$ $(= 6.875 \times 10^{-7}$ m)	**C1**
	$f = v / \lambda$	**C1**
	$= 3.0 \times 10^{8} / 6.875 \times 10^{-7}$ $= 4.4 \times 10^{14}$ Hz	**A1**
4(c)	the light / beams (have different frequencies so) are not coherent / do not have constant phase difference	**B1**

Question	Answer	Marks